

Badminton Action Classification Based on Human Skeleton Data Extracted by AlphaPose

Zhantu Liang
Faculty of Engineering, Science and
Technology,
Infrastructure University Kuala
Lumpur, Kajang, Malaysia
223923795@s.iukl.edu.my

Tadiwa Elisha Nyamasvisva *
Faculty of Engineering, Science and
Technology,
Infrastructure University Kuala
Lumpur, Kajang, Malaysia
tadiwa.elisha@iukl.edu.my

Abstract—In the realm of sports analytics, the precise classification of intricate movements is paramount. Badminton, a sport celebrated worldwide, is a testament to this with its swift and complex actions. While the domain of human action recognition has seen significant advancements, the specific exploration of badminton action classification using skeleton keypoint data remains a largely uncharted territory. This study bridges this research gap. Utilizing the AlphaPose framework, we harnessed skeleton keypoint data from selected badminton videos, offering a novel approach to understanding and categorizing the sport's nuanced movements. Our analysis showcased the superiority of the Long Short-Term Memory (LSTM) model, achieving an impressive 80% prediction accuracy, in contrast to the Convolutional Neural Network (CNN) model's 60%. This distinction emphasizes LSTM's adeptness at handling sequential data inherent to skeleton keypoints. In essence, this research not only underscores the untapped potential of skeleton keypoints in revolutionizing badminton analytics but also paves the way for future endeavors in this specialized domain.

Keywords—Badminton action classification, Skeleton keypoint data, AlphaPose, Long Short-Term Memory (LSTM)

I. INTRODUCTION

Badminton, a globally renowned sport, is characterized by its intricate movements and swift actions [1]. These actions, executed with precision, timing, and technique, hold immense significance for coaches, players, and sports scientists [2]. The capability to accurately understand and classify these actions can revolutionize training methodologies, enhance performance, and contribute to injury prevention.

The domain of human action recognition has seen remarkable progress, largely attributed to the advancements in deep learning and computer vision [3]. Among the available tools, AlphaPose has emerged as a notable solution for human pose estimation, proficiently extracting skeleton keypoint data from video sequences [4]. The utilization of skeleton keypoint data offers several compelling advantages. It inherently improves the model's resilience to occlusions, a frequent challenge in sports analytics. By focusing on key skeleton points, the training burden is significantly reduced. Furthermore, these keypoints, which directly reflect the human skeleton structure and posture, enhance the model's ability to learn human representations and understand body dynamics [5].

Despite the evident benefits of skeleton keypoint data, its application in the specific domain of badminton action classification remains surprisingly sparse. This paper seeks to address this research gap. By harnessing the capabilities of

AlphaPose, we extract skeleton data from a select set of badminton videos and employ advanced machine learning techniques to classify distinct badminton actions. Our research not only underscores the untapped potential of skeleton keypoint data in badminton analytics but also emphasizes the profound research significance and value it brings to the sport's domain.

II. THEORETICAL BACKGROUND

A. Foundations of Skeleton Keypoint Representation

Skeleton keypoints serve as pivotal landmarks on the human body, typically identified at major joints such as the elbows, knees, hips, and shoulders [6]. These keypoints, when connected, form a skeleton structure that provides a concise yet comprehensive representation of human posture and movement. This form of representation has gained prominence in computer vision and action recognition due to several inherent advantages. Firstly, skeleton keypoints filter out extraneous visual details, focusing solely on the core structure of human movement [7], [8]. This results in a reduced data dimensionality, making subsequent processing and analysis more efficient. Secondly, the skeleton representation remains consistent across various lighting conditions, backgrounds, and even clothing variations, ensuring that the essential movement patterns are consistently captured [9].

B. Deep Learning in Pose Estimation: The AlphaPose Framework

Pose estimation, which entails determining the position and orientation of specific body parts in visual data, has witnessed substantial advancements with the introduction of deep learning techniques, with AlphaPose emerging as a leading solution among numerous tools and frameworks in this realm [10]. Rooted in the foundation of deep convolutional neural networks, AlphaPose adopts a top-down approach for pose estimation, initially detecting each human in an image or video frame and, post detection, independently estimating the pose for each identified human region [11]. The essence of AlphaPose's prowess in extracting skeleton keypoints from images is the incorporation of the Regional Multi-person Pose Estimation (RMPE) framework [12], which houses several pivotal components: the Symmetric Spatial Transformer Network (SSTN) adept at managing spatial transformations within imprecise bounding boxes to extract high-quality human regions; the Parametric Pose Non-Maximum-Suppression (NMS) employed to eradicate redundant detections by comparing pose similarities through a novel pose distance metric, effectively differentiating

between unique and redundant poses; and the Pose-Guided Proposals Generator (PGPG) that augments training data by learning to generate bounding box proposals from varying poses, thereby enlarging and diversifying the training set to enhance the model's pose estimation accuracy [13], [14], [15]. These components synergistically operate, enabling AlphaPose to precisely identify and track skeleton keypoints within images and video sequences, overcoming challenges such as occlusions, multiple individuals, and rapid movements, making AlphaPose a preferred choice for those interested in extracting detailed skeleton data from visual mediums [16].

C. Techniques in Action Classification

Action classification, a cornerstone of human activity recognition, involves categorizing distinct human movements into predefined classes or actions. The process typically commences with feature extraction, where essential patterns or characteristics of the action are distilled from the raw data. Subsequent stages involve the training of classifiers using various machine learning algorithms. In recent years, deep learning, particularly recurrent neural networks (RNNs) like Long Short-Term Memory (LSTM) networks, has become a popular approach in this domain [17], [18]. LSTMs, with their memory cells, are designed to recognize and remember long-term dependencies in sequence data. This capability allows them to capture the intricate dynamics of actions over time, making them particularly adept at understanding the temporal progression of movements [19], [20]. Such attributes are crucial for sports like badminton where movements are rapid and sequential. Therefore, in this study, we have chosen to harness the power of LSTMs for insightful analyses of badminton player actions and strategies.

III. METHODOLOGY

In this section, we delve into the detailed procedures and techniques employed in our research, encompassing frame sampling, data processing, skeleton keypoint feature extraction, and the architecture of our classification model.

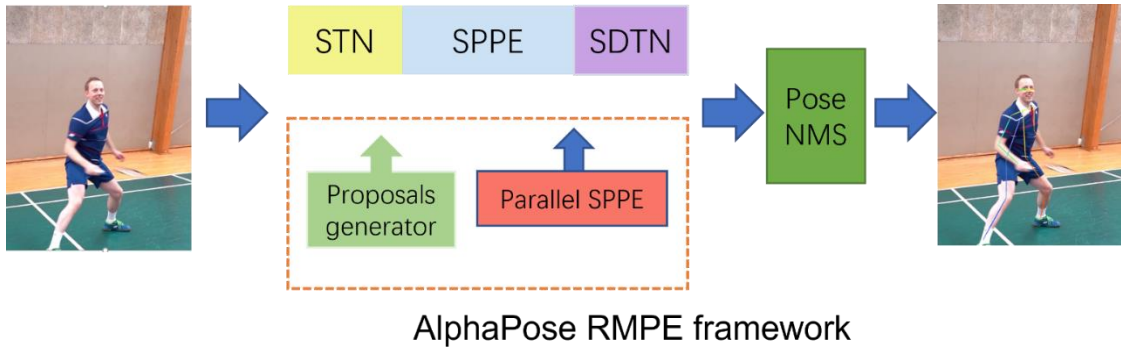


Fig. 1. Skeleton keypoint extracted from AlphaPose

C. Model Architecture

For our study on action classification using skeleton keypoints, we employed the Long Short-Term Memory (LSTM) network, a specialized recurrent neural network adept at handling sequence data. LSTMs are particularly suited for datasets like ours, where the temporal relationships between skeleton keypoints across frames are essential for accurate classification.

A. Frame Sampling

Frame sampling plays a pivotal role in video data processing, determining which frames are extracted for further analysis. Given the dynamic nature of badminton movements, it's essential to capture the most representative frames that encapsulate the essence of each action. Mathematically, the difference between two consecutive frames i and j can be represented as:

$$\mathcal{D}(i, j) = \sum_{x,y} |\mathcal{F}(i, x, y) - \mathcal{F}(j, x, y)| \quad (1)$$

where $\mathcal{D}(x, y)$ represents the difference between frames i and j , and $\mathcal{F}(i, x, y)$ denotes the pixel value of frame i at position x, y . If $\mathcal{D}(i, j)$ exceeds a certain threshold, frame j is considered a keyframe. This threshold can be determined

empirically or based on the specific requirements of the analysis. In our study, by employing the Frame Difference Method, we ensured that the most representative moments in badminton video sequences were selected for further analysis.

B. Skeleton Keypoint Feature Extraction

Utilizing the AlphaPose framework, as shown in Fig 1, we extracted skeleton keypoints from individual frames that were previously obtained from the video. For each of these frames, AlphaPose identifies human figures and outputs a set of skeleton keypoints, which represent the position and orientation of specific body parts. These keypoints are crucial for understanding and categorizing various badminton movements. The data is structured in a JSON format, with each keypoint characterized by a triplet of values: the x-coordinate (horizontal pixel position), the y-coordinate (vertical pixel position), and a confidence score that indicates the reliability of the detected keypoint. This spatial information, combined with the confidence metric, offers insights into the body's position in each frame and aids in filtering potential detection inaccuracies, enabling a comprehensive understanding of the badminton movements under study.

Given our dataset's limited size, it was crucial to design the network to be robust against overfitting. To achieve this, we structured our LSTM network with five layers. As shown in Fig 2, skeleton keypoints data is used as input. Each layer is tailored to handle time series data, with the input dimensions being adjusted to align with the keypoint data's dimensions. Upon training through the LSTM model, the output from the final time step is channeled to a fully connected layer. This subsequent layer translates the hidden states into predefined badminton action class, culminating in the classification process.

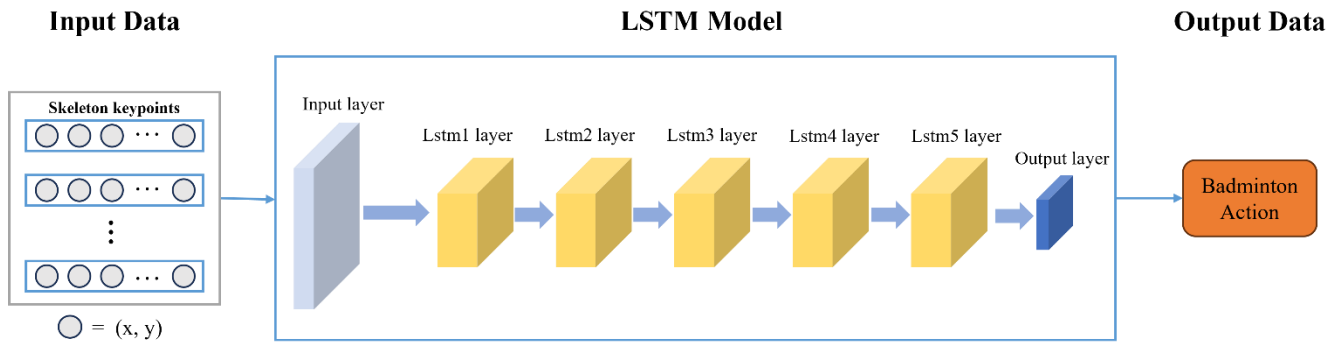


Fig. 2. LSTM network structure for badminton action classification

IV. EXPERIMENTS

A. Experimental platform and Dataset

In this paper, our research endeavors are powered by Google's Colab, a versatile, cloud-based computing environment that democratizes access to high-end computational resources by offering free usage of Graphics Processing Units (GPUs). Operating on a robust Linux-based system, Colab seamlessly integrates with various data science tools and libraries. For our specific experimental needs, we chose the PyTorch framework, a dynamic and flexible deep learning platform that is increasingly becoming the go-to choice for both academia and industry in the realm of artificial intelligence.

In the realm of sports analytics, particularly for badminton, there is a noticeable scarcity of comprehensive datasets. Addressing this need, our dataset for this research was sourced from Kaggle, a renowned platform for machine learning and data science projects. The dataset is tailor-made for the sport of badminton, with 450 videos in four different folders, each dedicated to a unique badminton action: backhand drive, backhand net shot, forehand clear, forehand drive.

To maintain the quality and consistency of the experiments, we carefully screened the videos in each folder. After a rigorous selection process, as shown in Table 1, we selected 100 videos qualified for training for each action, a total of 400 videos. We extract 10 frames from each video, for a total of 1000 frames per action, 4000 frames for the entire dataset. The frames are grouped by video, and each group of 10 frames is treated as a single sequence during training of the LSTM network. The data set is divided into training set, test set and verification set according to the ratio of 8:1:1 to ensure that the data distribution of model training and evaluation is balanced.

TABLE I. BADMINTON ACTION DATASET

Action	Number of Videos	Frames per Video	Total Frames
Backhand Drive	100	10	1000
Backhand Net Shot	100	10	1000
Forehand Clear	100	10	1000
Forehand Drive	100	10	1000
Total	400	10	4000

B. Analysis of experimental results

In this study, the LSTM model was trained throughout the regimen with an input size of 51, representing the 17 skeleton keypoints of the human body. The model consists of 128 hidden units, an output size of 4 (corresponding to the four actions), and is structured with 5 layers. Training was conducted using the cross-entropy loss function, optimized by the Adam optimizer at a learning rate of 0.001, over 110 epochs. The model utilizes multiple cyclic iterations to gradually optimize the parameters of the neural network during the training process, and is able to capture the temporal dependencies of the skeleton keypoints sequence. The nature of the LSTM enables it to remember previous time points' information, which is crucial in action recognition where the current state often depends on previous ones. The training process is shown in Fig 3. As the number of training epochs increased, there was a notable trend of the training accuracy improving, underscoring the model's ability to learn and adapt over time.

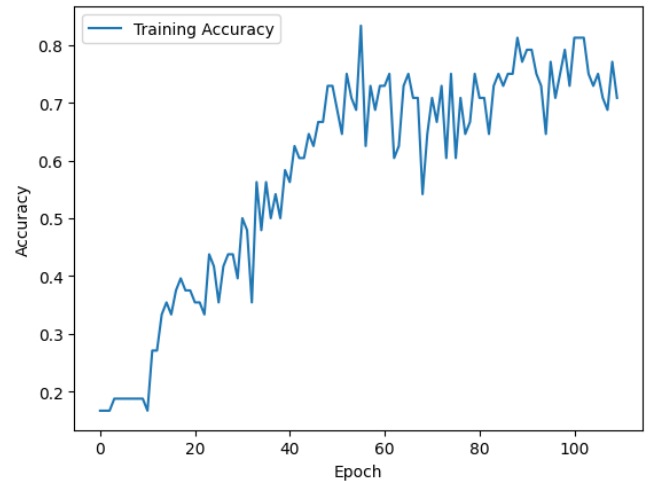


Fig. 3. LSTM train

The model demonstrated an impressive accuracy rate of 80% in its predictions. We employed a confusion matrix, as depicted in Fig 4, to elucidate the model's classification performance across different categories. A close examination of the matrix reveals a high degree of diagonal concentration, signifying correct predictions for most cases. There were only a few off-diagonal entries, indicating occasional misclassifications. Despite the constraints posed by the limited sample size, the model's ability to effectively classify sequences of skeletal keypoints stands out, underscoring the robustness of the LSTM approach in this context.

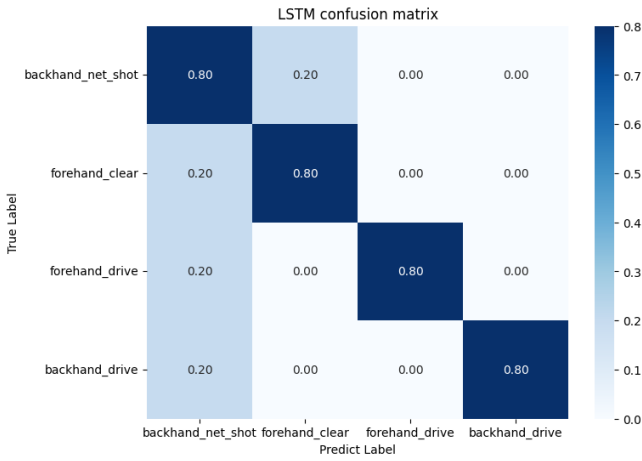


Fig. 4. LSTM confusion matrix

To compare with the performance of the LSTM model, we also designed a CNN neural network to classify the skeleton keypoint data. As shown in Fig 5, the classification results are also represented in the form of a confusion matrix.

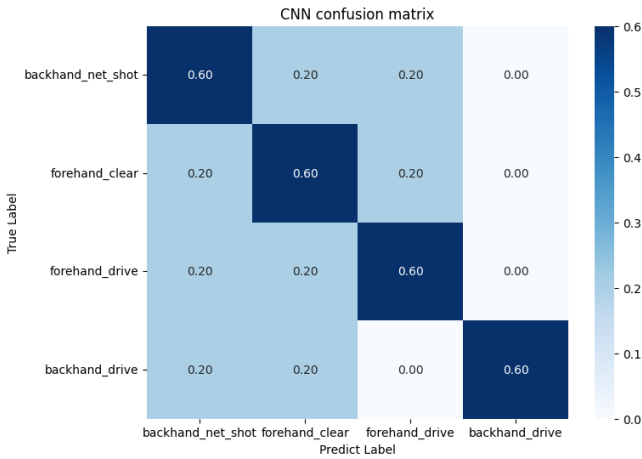


Fig. 5. CNN confusion matrix

In the quest to delineate the relative performance of various neural architectures on the dataset in question, we present the comparative accuracy rates in Table 2. The Convolutional Neural Network (CNN) procures an accuracy of merely 60%, thereby falling short in comparison to the LSTM model. A plausible explanation for this discernible disparity could be attributed to the intrinsic ability of the LSTM model to capture and leverage temporal dependencies present in sequence data. Such a capacity endows the LSTM with a distinct advantage when processing time-series data, such as skeleton keypoints. On the contrary, while CNNs excel in spatial feature extraction, they might not exhibit the same proficiency in comprehending the subtle temporal intricacies inherent in the dataset.

TABLE II. TEST ACCURACY OF LSTM AND CNN MODELS

Model	Test Accuracy (%)
LSTM	80
CNN	60

V. CONCLUSION

In conclusion, in this research, we delved into the niche area of badminton action recognition using skeleton keypoints data, an avenue less traveled in the domain of sports action analysis. By leveraging the AlphaPose framework, we efficiently extracted these keypoints from video frames, underscoring the potential of skeleton information in deciphering and categorizing intricate badminton movements. Our findings showcased the efficacy of LSTM and CNN models, with the LSTM model outperforming the latter, thus emphasizing its suitability for sequential data like skeleton keypoints. Notably, while our study illuminates the badminton action analysis landscape, offering insights for future researchers, it is essential to consider the constraints imposed by our limited dataset. Future researchers can improve classification accuracy and expand the applicability of badminton by expanding the dataset and improving the neural network architecture.

REFERENCES

- [1] Ghosh I, Ramamurthy S R, Chakma A, et al. Decoach: Deep learning-based coaching for badminton player assessment[J]. *Pervasive and Mobile Computing*, 2022, 83: 101608.
- [2] Adel B, Badran A, Elshami N E, et al. A Survey on Deep Learning Architectures in Human Activities Recognition Application in Sports Science, Healthcare, and Security[C]//*The International Conference on Innovations in Computing Research*. Cham: Springer International Publishing, 2022: 121-134.
- [3] Sharma V, Gupta M, Pandey A K, et al. A review of deep learning-based human activity recognition on benchmark video datasets[J]. *Applied Artificial Intelligence*, 2022, 36(1): 2093705.
- [4] Zhang J, Lim J, Kim M H, et al. WM-STGCN: A Novel Spatiotemporal Modeling Method for Parkinsonian Gait Recognition[J]. *Sensors*, 2023, 23(10): 4980.
- [5] Chen H, Feng R, Wu S, et al. 2D Human pose estimation: A survey[J]. *Multimedia Systems*, 2022: 1-24.
- [6] Kun E, Javan E M, Smith O, et al. The genetic architecture and evolution of the human skeletal form[J]. *Science*, 2023, 381(6655): eadf8009.
- [7] Wong C Y, Vergez L, Suleiman W. Vision-and Tactile-Based Continuous Multimodal Intention and Attention Recognition for Safer Physical Human-Robot Interaction[J]. *IEEE Transactions on Automation Science and Engineering*, 2023.
- [8] Guo T, Liu H, Chen Z, et al. Contrastive learning from extremely augmented skeleton sequences for self-supervised action recognition[C]//*Proceedings of the AAAI Conference on Artificial Intelligence*. 2022, 36(1): 762-770.
- [9] Su Y, Zhu H, Tan Y, et al. Prime: Privacy-preserving video anomaly detection via Motion Exemplar guidance[J]. *Knowledge-Based Systems*, 2023: 110872.
- [10] Gamra M B, Akhloufi M A. A review of deep learning techniques for 2D and 3D human pose estimation[J]. *Image and Vision Computing*, 2021, 114: 104282.
- [11] Fang H S, Li J, Tang H, et al. Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time[J]. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.
- [12] Fang H S, Xie S, Tai Y W, et al. Rmpe: Regional multi-person pose estimation[C]//*Proceedings of the IEEE international conference on computer vision*. 2017: 2334-2343.
- [13] Nguyen T D, Kresovic M. A survey of top-down approaches for human pose estimation[J]. *arXiv preprint arXiv:2202.02656*, 2022.
- [14] Hosang J, Benenson R, Schiele B. A convnet for non-maximum suppression[C]//*Pattern Recognition: 38th German Conference, GCPR 2016, Hannover, Germany, September 12-15, 2016, Proceedings 38*. Springer International Publishing, 2016: 192-204.
- [15] Patel M, Kalani N. A Comparative Analysis for Single Person and Multi Person Pose Estimation Using Deep Learning Algorithms[C]//*2021 International Conference on Circuits, Controls and Communications (CCUBE)*. IEEE, 2021: 1-5.

- [16] Fang H S, Li J, Tang H, et al. Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022.
- [17] Petković T, Petrović L, Marković I, et al. Human action prediction in collaborative environments based on shared-weight LSTMs with feature dimensionality reduction[J]. Applied Soft Computing, 2022, 126: 109245.
- [18] Senthilkumar N, Manimegalai M, Karpakam S, et al. Human action recognition based on spatial-temporal relational model and LSTM-CNN Framework[J]. Materials Today: Proceedings, 2022, 57: 2087-2091.
- [19] Saoudi E M, Jaafari J, Andaloussi S J. Advancing human action recognition: A hybrid approach using attention-based LSTM and 3D CNN[J]. Scientific African, 2023, 21: e01796.
- [20] Guan S, Lu H, Zhu L, et al. Afe-cnn: 3d skeleton-based action recognition with action feature enhancement[J]. Neurocomputing, 2022, 514: 256-267.